3.11 Polynomial Rings in Many Variables

Definition 3.11.1 (Polynomial ring in many variables)
Let A be a ring then the polynomial ring $A[X_1, ..., X_n]$ is the monoid ring $A[\mathbb{N}^n]$ (3.7.1) consisting of formal sums of monomials

$$f(X_1, ..., X_n) = \sum_{v \in \mathbb{N}^n} f_v X_1^{v_1} \cdots X_n^{v_n}$$

where $f_v \in A$ and only finitely many coefficients are non-zero. Addition is defined in the obvious way.

We may canonically regard A, $A[X_i]$ and $A[X_1, ..., X_i]$ as subrings in the obvious way.

Define $\deg(f, i)$ to be the maximal power of X_i with a non-zero coefficient.

Remark 3.11.2
It may be useful for certain induction arguments to write

$$A[X_1, ..., X_n] = A$$

when $n = 0$.

Proposition 3.11.3 (Evaluation Homomorphism)
$A[X_1, ..., X_n]$ satisfies the following universal property. Given any A-algebra B and points $(b_1, ..., b_n)$ there exists a ring homomorphism

$$\phi_b : A[X_1, ..., X_n] \to B$$

such that

$$\phi_b(X_i) = \phi(b_i)$$

given by

$$\phi_b\left(\sum_v a_v X_1^{v_1} \cdots X_n^{v_n}\right) = \sum_v i_B(a_v)\phi(b_1)^{v_1} \cdots \phi(b_n)^{v_n}$$

In other words it is an initial object in the category of n-pointed A-algebras. Furthermore

$$\mathrm{Im}(\phi_b) = A[b_1, ..., b_n]$$

Lemma 3.11.4 (Iterated polynomial ring)
Given $f \in A[X_1, ..., X_n]$ and let $N = \deg(f, n)$ then there exist unique polynomials $g_i \in A[X_1, ..., X_{n-1}]$ such that

$$f = \sum_{i=0}^{N} g_i X_n^i$$

in other words there is a canonical isomorphism

$$\psi : A[X_1, ..., X_{n-1}][X_n] \to A[X_1, ..., X_n]$$

under which $\deg(f) = \deg(\psi(f); n)$.

Proposition 3.11.5 (Homogenous grading)
Consider $R = k[X_1, ..., X_n]$ and $x \in k^n$. Then there is a direct sum of k-submodules

$$R = \bigoplus_{n \geq 0} R^{n,x}$$

where

$$R^{n,x} = \left\{ \sum_{|\alpha|_1 = n} \lambda_\alpha (X_1 - x_1)^{\alpha_1} \cdots (X_n - x_n)^{\alpha_n} \;\middle|\; \lambda_\alpha \in k \quad \alpha \in \mathbb{N}^n \right\}$$

and every $F \in R$ may be written uniquely as

$$F(X) = F(x) + F^{(1,x)}(X) + ... + F^{(n,x)}(X) + ...$$

with $F^{(n,x)} \in R^{n,x}$. Note that

$$\ker(\mathrm{ev}_x) =: M_x = \bigoplus_{n\geq 1} R^{n,x} = (X_1 - x_1, ..., X_n - x_n)$$

and

$$M_x^k = \bigoplus_{n\geq k} R^{n,x}$$

Finally there is a canonical isomorphism

$$k[X_1, ..., X_n]^{1,x} \cong M_x/M_x^2$$

Proof. By Proposition (...) there is a k-algebra homomorphism $\rho_x : R \to R$ given by $X_i \to X_i + x_i$. It is an isomorphism with two-sided inverse ρ_{-x}. Let $F \in R$ then

$$\rho_x(F) = \sum_{n=0}^{\infty} \left(\sum_{|\alpha|_1 = n} \lambda_\alpha X_1^{\alpha_1} \cdots X_n^{\alpha_n} \right)$$

whence applying ρ_{-x}

$$\begin{aligned} F(X) &= \sum_{n=0}^{\infty} F^{(n)}(X) \\ F^{(n)}(X) &= \sum_{|\alpha|_1 = n} \lambda_\alpha (X_1 - x_1)^{\alpha_1} \cdots (X_n - x_n)^{\alpha_n} \end{aligned}$$

as required. The coefficients λ_α are seen to be uniquely determined by applying ρ_x. Therefore the internal sum is direct. Finally evaluate at x to find $F^{(0)} = F(x)$. The statement regarding M_x is straightforward. And because $R^{n,x} \cdot R^{m,x} \subseteq R^{n+m,x}$ the statement regarding M_x^k follows by induction. ∎

Definition 3.11.6 (Projection to linear terms)
Given $\mathfrak{a} \lhd k[X_1, ..., X_n]$ and $x \in k^n$ define

$$\mathfrak{a}^{(i,x)} = \left\{ F^{(i,x)} \;\middle|\; F \in \mathfrak{a} \right\}$$

The following is useful

Lemma 3.11.7
Let A be a k-algebra and $F \in k[X_1, ..., X_n]$ and $G_1, ..., G_n \in k[Y_1, ..., Y_m]$ polynomials. For $\lambda_1, ..., \lambda_m \in A$ we have

$$F(G_1, ..., G_n)(\lambda_1, ..., \lambda_m) = F\Big(G_1(\lambda_1, ..., \lambda_m), ..., G_{n(\lambda_1, ..., \lambda_m)}\Big)$$